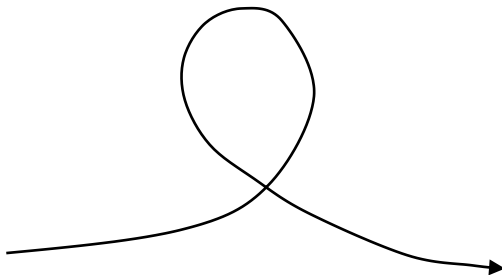


Question			Solution	Marks
8	(a)		• photon gives energy to electron (in an inner shell) or electron (in an inner shell) absorbs a photon • electron moves (from lower) to higher energy level • energy (of photon) is equal to difference in energy levels • electron de-excites giving off photon (of same energy) • photons emitted in all directions	B1 B1 B1
	(b)	(i)	X-ray photon produced when electron is decelerated larger acceleration results in larger photon energy continuous range of accelerations so continuous spectrum of wavelengths/frequencies	B1 B1 B1
		(ii)	electron in (inner shell of) target atom is excited (on collision) electron de-excites causing emission of a photon discrete energy levels so discrete photon wavelengths	B1 B1 B1

Question		Solution	Marks
	(a)	$E = hf$ $hf = kB$ $(6.63 \times 10^{-34})(42.58 \times 10^6) = k(1.0)$ $k = 2.82 \times 10^{-26} \text{ J T}^{-1}$	C1 C1 A1
	(b) (i)	Straight line that cuts through origin So RF proportional to B	M1 A1
	(ii)	$K = \text{gradient} \times \text{Planck's Constant}$	
	(iii)	Scattering of points about the best fit line	B1
	(c)	Varying flux density B Photon absorbed (or emitted) only when energy = kB OR Photon absorbed (or emitted) proportional to B Only specific areas will absorb (or emit)	B1 B1 (B1)
	(d)	Wavelength of X-ray = 10^{-10} m $\text{Energy of X-ray} = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{10^{-10}(1.6 \times 10^{-19})} = 1.2 \times 10^4 \text{ eV}$ X-ray energy is larger than molecular bonds, in the order of 1 eV.	B1 M1 A1
	(e) (i)	1. forces act through the same line or forces are horizontal forces are equal (in magnitude) and opposite (in direction)	B1 B1
		2. They are parallel to the field	B1
	(ii)	$mg \times (\Delta)L = B \times (\Delta)l \times L \times x$ $2.5 \times 10^{-3} \times 9.81 \times 2.6 \times 10^{-2} = B \times 1.2 \times 0.85 \times 10^{-2} \times 5.6 \times 10^{-2}$ $B = 1.1 \text{ T}$	C1 C1 A1
	(f)	Spiral with radius decreasing towards the top of the field Anticlockwise direction 	M1 A1